Arboreal height of integer-interlaced cubic squares

Sam Vaseghi

Wisehouse Science Communication Lab, Wisehouse AB,
Ballingslöv, Sweden

Corresponding author: sam.vaseghi@wisehouse-publishing.com
ORCID: 0009-0006-4105-1949

29 October 2026

Abstract

The arboreal height of a totally real algebraic integer is the least height of a finite rooted tree having it as an adjacency eigenvalue. Let $\alpha = \lambda^2$ be a totally positive algebraic integer of degree three, and suppose that each open gap between consecutive conjugates of α contains an integer. We prove that λ has arboreal height at most three. Together with Schaeffer's height-two criterion, this gives exact height two when the minimal polynomial of α has a root modulo two, and exact height three otherwise. The proof uses interpolation to express additive inverses as finite nonnegative integer combinations of star resolvents. Salez's affine identity supplies inverses at the next branching level, where a cubic polynomial identity completes the construction. All intermediate branches are invertible at the prescribed eigenvalue. A dilation corollary determines the height of every sufficiently large integer multiple of a square root of an arbitrary totally positive cubic algebraic integer.

Keywords: tree eigenvalues; arboreal height; totally real algebraic integers; integer interlacing; rooted resolvents.

Mathematics Subject Classification (2020): Primary 11R04; Secondary 05C50, 11R16.

1 Minimum height and the cubic case

For a finite rooted tree (T, o), its height is the largest distance from o to a vertex of T. We write $h(\lambda)$ for the least height among finite rooted trees whose adjacency matrix has eigenvalue λ. The root is free to vary, so this is also the least radius among all such trees. Throughout the paper, trees are finite, simple and unweighted.

Salez proved that every totally real algebraic integer is an eigenvalue of a tree [1]. Once existence is known, the minimum height asks how many successive branching levels are needed. A star has height one and nonzero eigenvalues whose squares are integers. At height two, the branches attached to the root are stars, and their contributions lead to the fractions $1/(\lambda^2 - j)$ with j a nonnegative integer. Schaeffer used this arithmetic description to characterize height two for cubic squares [2, Theorem 10].

Let α be a totally positive algebraic integer of degree three, with minimal polynomial

$$F(x) = x^3 - Sx^2 + Px - Q \in \mathbb{Z}[x] \tag{1}$$

and conjugates

$$0 < \alpha_1 < \alpha_2 < \alpha_3.$$

We say here that the conjugates are *integer-interlaced* if both (α_1, α_2) and (α_2, α_3) contain an integer. Since F is irreducible of degree three, none of its roots is an integer. Either square root of any α_i is a totally real algebraic integer: it is a root of the monic integral polynomial $F(x^2)$, all of whose roots are real. The degree of the square root may be three or six.

Theorem 1. Let $\alpha = \lambda^2$ have the properties above, including integer interlacing. Then

$$h(\lambda) = \begin{cases} 2, & F \text{ has a root in } \mathbb{F}_2, \\ 3, & F \text{ has no root in } \mathbb{F}_2. \end{cases} \tag{2}$$

Schaeffer's theorem supplies the first case and excludes height two in the second [2]. The addition proved here is the ceiling $h(\lambda) \leq 3$ throughout the integer-interlaced class. In particular, when the parity condition fails, one further branching level always suffices.

Dubickas's general height theorem gives the bound nine when $[\mathbb{Q}(\lambda^2) : \mathbb{Q}] = 3$ [3, Theorem 3]. Appendix A explains how information from interlacing changes the starting level of his proof and compares the resulting bounds with the direct height-three argument below.

Explicit diameter-six constructions provide another relevant comparison. Wang and coauthors [4] and Xi and Wang [5, Theorems 2.1–2.2 and Corollaries 2.1–2.2] give characteristic-polynomial formulas that include cubic factors in the squared eigenvalue. Using these formulas for an arbitrary prescribed cubic requires positive integer parameters satisfying the coefficient equations. Their formulas and conditional inverse calculations do not establish that parameter choice for every cubic in Theorem 1. The construction below produces finite nonnegative branch counts directly from the interlacing assumption. The example in Section 6 is a boundary specialization of the established Xi–Wang template and is included to illustrate the result.

Beezer's recursive construction instead produces trees with exactly one more distinct eigenvalue than their diameter [6, Theorem 2.2]. This spectral restriction prevents that family from realizing every cubic in Theorem 1 at radius three: Section 6 gives an explicit cubic for which no tree satisfying Beezer's distinct-eigenvalue condition can have diameter at most six. His term *minimal* refers to the number of distinct eigenvalues for a given diameter, whereas the minimum considered here is the height needed for a prescribed eigenvalue.

The proof follows the branching calculation. We first describe the arithmetic quantities represented by stars and by trees of height two. Interpolation then makes subtraction possible within their nonnegative integer sums. A further step for constructing additive inverses, already present in Salez's argument, puts α in the collection needed for a height-three tree.

2 Arithmetic quantities represented by branches

Fix one root α of F and a real λ with $\lambda^2 = \alpha$. For $j \geq 0$, let S_j be the star with j leaves, rooted at its center; S_0 is a single vertex. For any rooted tree (B, b) for which $zI - A_B$ is invertible, define its root resolvent by

$$m_B(z) = e_b^{\mathsf{T}}(zI - A_B)^{-1}e_b,$$

where e_b is the coordinate vector of the root. To see how this quantity determines an eigenvalue, let T be formed by joining a new vertex to the roots of disjoint trees $B_1, \ldots, B_k$. If each $\lambda I - A_{B_i}$ is invertible, block elimination gives

$$\det(\lambda I - A_T) = \left(\prod_{i=1}^{k} \det(\lambda I - A_{B_i})\right)\left(\lambda - \sum_{i=1}^{k} m_{B_i}(\lambda)\right).$$

The product is nonzero, so λ is an eigenvalue of T exactly when the last factor vanishes. This last factor is the root Schur complement. Eliminating the leaves of a star gives

$$m_{S_j}(\lambda) = \frac{\lambda}{\alpha - j}. \tag{3}$$

These inverses exist because $\lambda \neq 0$ and α is not an integer. Formula (3) also holds for $j = 0$, when its value is $1/\lambda$.

Suppose a new root is joined to the centers of a_j disjoint copies of S_j, with nonnegative integers a_j and only finitely many nonzero. The root Schur complement at λ is

$$\lambda - \sum_{j \geq 0} a_j m_{S_j}(\lambda) = \lambda \left(1 - \sum_{j \geq 0} \frac{a_j}{\alpha - j} \right). \tag{4}$$

Thus the relevant arithmetic set is

$$r_j = \frac{1}{\alpha - j}, \qquad \Gamma = \left\{ \sum_{j \geq 0} a_j r_j : a_j \in \mathbb{Z}_{\geq 0},\ a_j = 0 \text{ eventually} \right\}. \tag{5}$$

The coefficients in (5) count copies of branches. A signed identity in the field $\mathbb{Q}(\alpha)$ would not yet specify a tree, because negative copies are unavailable. The next section shows that interlacing lets every negative generator be replaced by a finite sum with nonnegative integer coefficients.

For $s \in \Gamma$, choose a representation in (5) and denote the corresponding tree in (4) by $B(s)$. The choice need not be unique; any one finite representation is sufficient. This tree has height at most two. When $s \neq 1$, equation (4) shows that $\lambda I - A_{B(s)}$ is invertible and

$$\lambda m_{B(s)}(\lambda) = g(s), \qquad g(s) = \frac{1}{1 - s}. \tag{6}$$

Consequently, a finite identity

$$\alpha = \sum_{\ell=1}^{N} g(s_\ell), \qquad s_\ell \in \Gamma \setminus \{1\}, \tag{7}$$

will provide the branches of a height-three tree. The final root Schur complement will be $\lambda - \alpha/\lambda = 0$. Our algebraic task is therefore to obtain (7) while keeping every intermediate denominator nonzero.

Figure 1 follows the two attachments that lead from stars to the final tree. The representation of each s_ℓ fixes the stars inside its branch, while equation (7) determines how these branches contribute at the final root.

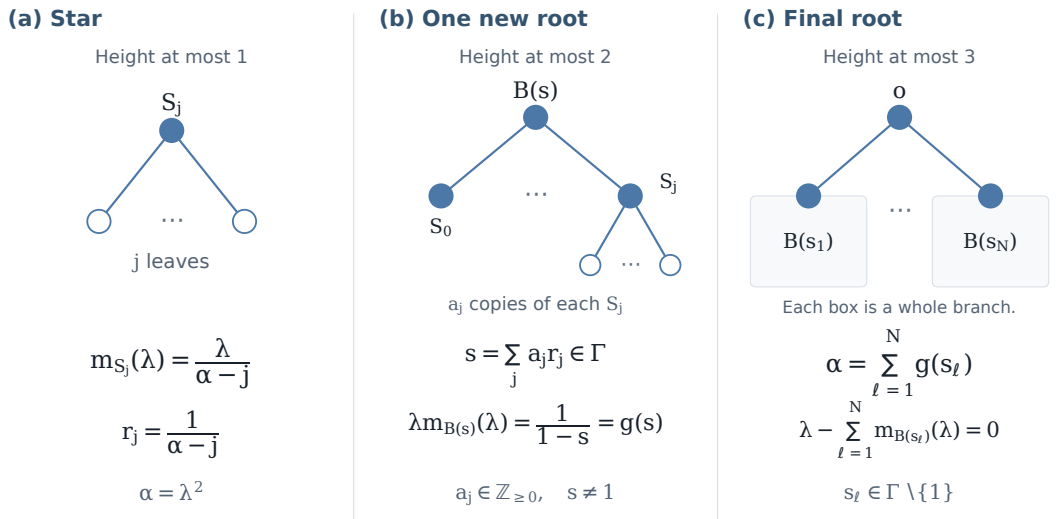


Figure 1: Arithmetic contributions at successive branching levels. Copies are schematic; their finite nonnegative multiplicities are specified by the representations in (5) and (7). Each branch $B(s_\ell)$ is invertible at λ when $s_\ell \neq 1$, and $\sum_\ell g(s_\ell) = \alpha$ makes the final root Schur complement vanish. Drawn from explicit graph definitions using deterministic Python/Matplotlib code prepared with AI assistance; details are given in the AI declaration.

3 Interlacing and nonnegative integer sums

We will construct additive inverses of the quantities $g(s)$ at the next branching level. This will use two properties of Γ: closure under subtraction and the presence of an integer $M > 1$.

Lemma 2. Under the hypotheses of Theorem 1, Γ is an additive group and contains an integer $M > 1$.

Proof. Choose integers

$$t_0 = 0 < \alpha_1 < t_1 < \alpha_2 < t_2 < \alpha_3 < t_3$$

and put $G(x) = \prod_{i=0}^{3}(x - t_i)$. Partial fractions give

$$\frac{F(x)}{G(x)} = \sum_{i=0}^{3} \frac{c_i}{x - t_i}, \qquad c_i = \frac{F(t_i)}{G'(t_i)} > 0. \tag{8}$$

The signs in the quotient agree at each t_i: the numbers of roots of F and of other poles of G lying to its right are the same. Evaluating (8) at α therefore gives a positive rational zero relation

$$\sum_{i=0}^{3} c_i r_{t_i} = 0. \tag{9}$$

Any three of these four resolvents form a rational basis of $\mathbb{Q}(\alpha)$. To verify independence, multiply a rational linear relation among three resolvents by their three denominators. The result is a polynomial of degree at most two vanishing at α, so it is the zero polynomial. Evaluating that polynomial at each of the three distinct poles makes the corresponding coefficient zero.

Now fix any $j \geq 0$. Express $-r_j$ as a rational linear combination of $r_{t_0}, r_{t_1}, r_{t_2}$, and give r_{t_3} coefficient zero. Adding a sufficiently large nonnegative rational multiple of (9) makes all four coefficients nonnegative. Clear their denominators to obtain

$$-Dr_j = \sum_{i=0}^{3} b_i r_{t_i}, \qquad D \in \mathbb{Z}_{\geq 1}, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

It follows that

$$-r_j = (D-1)r_j + \sum_{i=0}^{3} b_i r_{t_i} \in \Gamma. \tag{10}$$

This identity uses finitely many nonnegative integer coefficients. Repeating it for the generators in any given finite sum proves that Γ is closed under additive inverses.

To obtain an integer in Γ, retain only the first three poles and set $K(x) = \prod_{i=0}^{2}(x - t_i)$. Polynomial division and partial fractions give

$$\frac{F(x)}{K(x)} = 1 + \sum_{i=0}^{2} \frac{F(t_i)}{K'(t_i)(x - t_i)}.$$

Evaluation at α yields

$$1 = \sum_{i=0}^{2} d_i r_{t_i}, \qquad d_i = -\frac{F(t_i)}{K'(t_i)} > 0. \tag{11}$$

The signs are again determined by interlacing. Clearing the positive rational coefficients in (11) produces a positive integer in Γ; taking a positive multiple if necessary gives $M > 1$. $\square$

The interpolation in (8)–(11) uses the partial-fraction method of Salez and Schaeffer [1, Lemma 2; 2, Proposition 5]. Equation (10) spells out its consequence for every generator of the interlacing monoid. In particular, the group statement does not replace branch counts by arbitrary signed integers: it gives the nonnegative integer replacements that the next construction will use.

4 One further branching level

Let $\mathcal{H}$ be the additive monoid generated by the numbers $g(s)$ for $s \in \Gamma \setminus \{1\}$, where g is defined in (6). Thus its elements are finite sums with nonnegative integer multiplicities. We need $\alpha \in \mathcal{H}$ in order to obtain (7).

For the integer $M > 1$ from Lemma 2 and any $s \in \Gamma \setminus \{1\}$, put

$$t = M - (M-1)s.$$

The group property places t in Γ, and $1 - t = -(M-1)(1-s)$ shows that $t \neq 1$. Hence

$$(M-1)g(M - (M-1)s) = -g(s). \tag{12}$$

This is Salez's affine identity for additive inverses [1, §4, Step 2], applied to the fixed branching level represented by Γ. Its positive integer multiplier makes $-g(s)$ a permitted sum of generators. Therefore $\mathcal{H}$ is an additive group.

Since $g(0) = 1$, this group contains the integers. It also contains Γ, as follows from

$$g(r_{j-1}) = 1 + r_j \quad (j \geq 1), \qquad g(-r_1) = 1 - r_0. \tag{13}$$

Here r_{j-1} and $-r_1$ belong to Γ. The first argument could equal one only if $\alpha = j$, and the second only if $\alpha = 0$; neither is possible. Subtracting the integer terms in (13) therefore places every r_j in $\mathcal{H}$.

We have shown that $\mathcal{H}$ contains both $\mathbb{Z}$ and Γ. It therefore remains to express α as an integer plus an integer combination of the resolvents r_j. For a cubic, subtracting the following two monic quadratic quotients cancels their quadratic terms and leaves a linear expression in x:

$$\frac{F(x) - F(1)}{x - 1} = x^2 + (1 - S)x + (1 - S + P), \qquad \frac{F(x) - F(0)}{x} = x^2 - Sx + P.$$

Their difference is $x + 1 - S$. Setting $x = \alpha$ gives

$$F(1)r_1 - F(0)r_0 = -\alpha + S - 1. \tag{14}$$

Rearranging (14) gives the required expression, so $\alpha \in \mathcal{H}$.

For an explicit final nonnegative expression, use $F(0) = -Q$, $F(1) = 1 - S + P - Q$ and (13) to rewrite (14) as

$$\alpha = (P - 2Q)g(0) + Qg(-r_1) - F(1)g(r_0).$$

Write $c_+ = \max(c, 0)$ for an integer c. Since $Q = \alpha_1\alpha_2\alpha_3 > 0$, equation (12) gives

$$\begin{aligned} \alpha = {} & (P - 2Q)_+ g(0) + (2Q - P)_+ (M - 1)g(M) + Qg(-r_1) \\ & + (-F(1))_+ g(r_0) + F(1)_+ (M - 1)g\big(M - (M - 1)r_0\big). \end{aligned} \tag{15}$$

Every coefficient is a nonnegative integer. Every argument belongs to Γ by Lemma 2. The arguments 0 and $M > 1$ differ from one; $-r_1 = 1$ would give $\alpha = 0$, while either $r_0 = 1$ or $M - (M - 1)r_0 = 1$ would give $\alpha = 1$. Thus (15), with terms repeated according to their coefficients, is exactly a finite identity of the form (7).

Choose a finite nonnegative representation supplied by Lemma 2 for each argument that occurs in (15), and construct its tree $B(s_\ell)$ as in Section 2. Attach these trees by their roots to one new vertex o. All child stars are invertible at λ, and (6) proves that each entire branch $B(s_\ell)$ is invertible there. The root Schur complement of the resulting tree T is

$$\lambda - \sum_{\ell=1}^{N} m_{B(s_\ell)}(\lambda) = \lambda - \frac{1}{\lambda} \sum_{\ell=1}^{N} g(s_\ell) = \lambda - \frac{\alpha}{\lambda} = 0. \tag{16}$$

Therefore $\lambda I - A_T$ is singular. The tree has height at most three, and its branch counts are finite nonnegative integers. Only the final Schur complement in (16) vanishes; no cancellation of a singular intermediate branch is needed. This proves the upper bound in Theorem 1.

5 Exact height and integer dilations

We finish the proof of Theorem 1 by determining when height two is possible. Every height-two rooted tree consists of stars attached to its root. Since none of those stars has eigenvalue λ, equation (4) shows that such a tree has eigenvalue λ precisely when $1 \in \Gamma$. Under integer interlacing, Schaeffer's Theorem 10 states that this happens precisely when F has a root modulo two [2]. The hypothesis is the same as his left-interlacing condition: the three nonnegative integers $0, t_1, t_2$ lie successively before the three positive conjugates.

The necessary parity condition can also be seen directly from the branch equation. If F has no root modulo two, its cubic reduction $\overline{F}$ is irreducible over $\mathbb{F}_2$. Reduction sends α to an element a of degree three in the field $\mathbb{F}_2[x]/(\overline{F})$. Every denominator $\alpha - j$ remains nonzero, so reduction extends to the fractions in (5). An identity $1 \in \Gamma$ would reduce to

$$1 = \frac{b_0}{a} + \frac{b_1}{a-1}, \qquad b_0, b_1 \in \mathbb{F}_2.$$

Multiplication by $a(a-1)$ would make a a root of a monic quadratic over $\mathbb{F}_2$, contradicting its degree. This proves the exclusion of height two in the rootless case; the sufficiency when a root exists is Schaeffer's result.

Height zero gives only the eigenvalue zero, and a height-one tree is a star, whose nonzero eigenvalues have integer squares. Neither can realize a degree-three α. Combining this observation, Schaeffer's criterion and (16) proves (2). □

Integer dilation brings every totally positive cubic square within the theorem after a sufficient enlargement of its conjugate gaps.

Corollary 3. Let $\alpha = \lambda^2$ be any totally positive algebraic integer of degree three, with minimal polynomial (1), and set

$$\delta = \min\{\alpha_2 - \alpha_1, \alpha_3 - \alpha_2\} > 0.$$

For every nonzero integer d with $d^2\delta > 1$,

$$h(d\lambda) = \begin{cases} 2, & d \text{ is even, or } F \text{ has a root modulo 2,} \\ 3, & d \text{ is odd and } F \text{ has no root modulo 2.} \end{cases} \tag{17}$$

If α is already integer-interlaced, (17) holds for every nonzero integer d.

Proof. The conjugate gaps of $d^2\alpha$ have length at least $d^2\delta > 1$, so each contains an integer. The minimal polynomial of $d^2\alpha$ is

$$F_d(x) = d^6 F(x/d^2) = x^3 - Sd^2x^2 + Pd^4x - Qd^6. \tag{18}$$

It is monic and integral, and remains irreducible because $\mathbb{Q}(d^2\alpha) = \mathbb{Q}(\alpha)$. For odd d, its reduction modulo two equals that of F; for even d, it is x^3. Theorem 1 gives (17). If the original gaps already contain integers, multiplying those integers by d^2 supplies interlacing for every nonzero d. □

The scaling observation and the height-two conclusions belong to Schaeffer's framework [2, Theorems 10 and 12]; his Example 15 also exhibits a height-three value whose double has height two. The additional consequence here is the height-three ceiling for every sufficiently large odd dilation in the rootless cubic case. The threshold $d^2\delta > 1$ is sufficient and need not be the first dilation at which interlacing occurs.

6 A fifteen-vertex realization

Consider

$$F(x) = x^3 - 6x^2 + 9x - 1. \tag{19}$$

The signs at $0, 1, 2, 3, 4$ place its three roots in $(0, 1)$, $(2, 3)$ and $(3, 4)$. Modulo two it becomes $x^3 + x + 1$, which has no root and is therefore irreducible. Thus F is irreducible over $\mathbb{Q}$, its roots are totally positive and integer-interlaced, and Theorem 1 gives height exactly three for each real λ satisfying $F(\lambda^2) = 0$.

A small tree makes the construction visible. Start with a root o and attach three copies of the two-vertex path, each by one endpoint. Add a fourth branch whose root is joined to the centers of a two-leaf star and a three-leaf star. This branch has eight vertices, so the whole tree has $1 + 3 \cdot 2 + 8 = 15$ vertices and height three from o, as shown in Figure 2. Its diameter is five and its radius is three. Each two-vertex branch is $B(r_0)$, formed by attaching a new root to the single vertex S_0; the fourth branch is $B(r_2 + r_3)$.

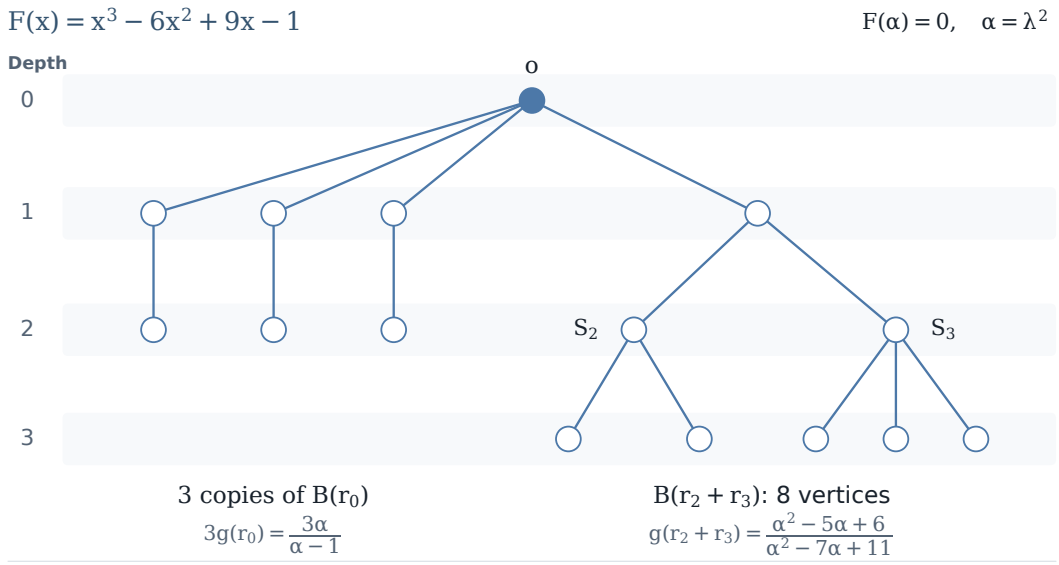


Figure 2: The tree realizing the roots of $F(\lambda^2) = 0$ for the cubic in (19). The levels contain $1, 4, 5, 5$ vertices. The three two-vertex branches each contribute $g(r_0) = \alpha/(\alpha - 1)$ to the normalized root equation; the eight-vertex branch contributes $g(r_2 + r_3)$. Drawn from explicit graph definitions using deterministic Python/Matplotlib code prepared with AI assistance; details are given in the AI declaration.

Writing $x = \lambda^2$, its root equation is

$$x = \frac{3x}{x - 1} + \frac{x^2 - 5x + 6}{x^2 - 7x + 11}. \tag{20}$$

The first term is $3g(r_0)$, the contribution of the three path branches. The second is $g(r_2 + r_3)$ for the eight-vertex branch. On multiplying (20) by $(x - 1)(x^2 - 7x + 11)$, the difference between its two sides becomes $(x - 6)F(x)$. Neither denominator can vanish at a root of the irreducible cubic F. The same calculation, retaining the branch determinants, gives the complete characteristic polynomial

$$\chi_T(z) = z^3(z^2 - 1)^2(z^2 - 6)F(z^2). \tag{21}$$

This tree is a boundary specialization of the Xi–Wang rooted-tree template [5]; Appendix B gives the parameter choice and re-rooting. Its structure is inherited from that established template, and the example illustrates the exact-height conclusion.

Comparison with Beezer's minimal trees

The same cubic separates this realization problem from Beezer's minimal trees [6]. Suppose a tree U of diameter at most six had exactly diameter-plus-one distinct eigenvalues and contained an eigenvalue λ with $F(\lambda^2) = 0$. Conjugacy and bipartite symmetry force all six roots of $F(z^2)$ into its spectrum. There is room for at most one additional distinct eigenvalue, which could only be zero. Writing k for the multiplicity of zero, the characteristic polynomial has the form $z^k P(z^2)$ with integral P. Irreducibility of F then gives $\chi_U(z) = z^k F(z^2)^m$. The largest eigenvalue of a connected tree is simple, so $m = 1$.

The coefficient -6 in (19) now says that U has six edges and hence seven vertices. The coefficient -1 says that it has exactly one matching of size three: in a tree, the coefficient of z^{n-2j} is $(-1)^j$ times the number of j-edge matchings, by the determinant expansion. But a three-edge matching in a connected seven-vertex tree leaves one vertex u unmatched. Choose a neighbor v of u and let vw be the matching edge at v. Replacing vw by uv gives a second three-edge matching, a contradiction. Thus Beezer's minimal-tree family cannot supply a universal radius-three realization for the class considered here.

The theorem controls height while allowing the number of branches to depend on the denominators cleared in Lemma 2. It gives no general small-order realization or bound optimal in the number of vertices. Cubics whose conjugate gaps do not contain integers remain outside Theorem 1, although Corollary 3 determines the heights of their sufficiently large integer dilations.

A Comparison with the signed-power construction

Dubickas's general theorem gives the bound nine for cubic squares [3, Theorem 3]. The bounds eight and seven obtained below are deductions from his proof with different starting information, rather than bounds stated in that paper.

For comparison with the general height argument, put $\beta = 1/\alpha$ and $f(s) = \beta/(1-s)$. In Dubickas's notation, the level-k additive semigroup is $L_k = F_\beta(k)$ [3]. It starts with $L_1 = \mathbb{Z}_{\geq 0}\beta$, and the next level is generated by the values $f(s)$ at the previous level, excluding $s = 1$. The levels are nested and closed under addition. In particular, $L_2 = \Gamma$. The normalizations are related by $f = \beta g$ and $L_3 = \beta\mathcal{H}$, so $1 \in L_3$ is exactly $\alpha \in \mathcal{H}$. More generally, $1 \in L_k$ yields a tree of height at most k. If 1 occurs at an earlier level, that earlier bound already suffices.

Interlacing supplies $M > 1$ in L_2 through the established identity (11). Thus $(M-1)f(M) = -\beta \in L_3$. Starting with both signs of β at level three, the induction in [3, Lemma 5] places both signs of β^2 and β^3 at levels five and eight. The identity $1 = S\beta - P\beta^2 + Q\beta^3$ then gives height at most eight. Lemma 2 instead places both signs of β in L_2, so the same induction gives levels four and seven and hence height at most seven. This latter comparison uses our group deduction in addition to the cited induction. Equations (12)–(16) reach height three by working with the next-level group and the cubic quotient identity, without generating successive signed powers through that induction.

B The Xi–Wang template for the example

The tree in Section 6 is a boundary specialization of the Xi–Wang rooted-tree template [5]. In their notation, $T(a, b)$ joins a new root to the centers of a copies of S_b, and $T(a, b, c)$ joins a new root to the roots of a copies of $T(b, c)$. The operation $\bullet$ identifies the roots of its factors. The unrooted tree is

$$T(1, 2) \bullet T(1, 3) \bullet T(1, 3, 1).$$

Here the first parameter of $T(1, 3, 1)$ is the boundary value $r = 1$ of their deeper branch family. Re-rooting at the root of its attached copy of $T(3, 1)$ gives the description in Section 6 and Figure 2.

Statements and declarations

Funding. No external funding has been declared for this work.

Competing interests. The author holds an ownership interest in Wisehouse AB, which is also the author's affiliation. The manuscript does not evaluate a commercial product or service.

Author contributions. S.V. is the sole author and is responsible for the research, manuscript and decision to submit. The role of AI assistance is described below.

Data and code availability. No empirical datasets were generated or analysed. The complete mathematical arguments are contained in the article. Deterministic figure-generation code and exact checks of the fifteen-vertex example are included in the accompanying source archive. The earlier manuscript and source archive are available at `https://doi.org/10.5281/zenodo.23479534`.

Ethics, consent to participate and consent to publish. Not applicable.

AI assistance. AI assisted the formulation of candidate arguments, literature comparison, proof development, algebraic calculations, manuscript drafting and revision, and preparation of the deterministic code for Figures 1 and 2. The author's designation for this assistance is OpenAI Astra Ultra; exact model-build identifiers and a complete prompt log were not recorded. The original manuscript and figure sources were prepared on 19 October 2026, and the manuscript was revised on 29 October 2026. The tasks concerned the interpolation identities, the cubic quotient calculation, intermediate denominators, source attribution and exposition. Checks included exact algebraic calculations, comparison with the cited sources, and integer-arithmetic verification of the example's characteristic polynomial. Separate AI agents provided internal assessments of the proof and exposition; these are not external peer review. The author is responsible for the mathematical claims, references and final text.

References

[1] Justin Salez, *Every totally real algebraic integer is a tree eigenvalue*, Journal of Combinatorial Theory, Series B **111** (2015), 249–256. https://doi.org/10.1016/j.jctb.2014.09.001. Preprint, arXiv:1302.4423v2.

[2] George J. Schaeffer, *Totally real algebraic integers of arboreal height two*, arXiv:2111.14256v1 (2021). https://arxiv.org/abs/2111.14256v1.

[3] Artūras Dubickas, *An Upper Bound for the Height of a Tree with a Given Eigenvalue*, Combinatorica **44** (2024), 299–310. https://doi.org/10.1007/s00493-023-00071-2. Author-linked journal text.

[4] Ligong Wang, Hajo Broersma, Cornelis Hoede, Xueliang Li and Georg Still, *Integral trees of diameter 6*, Discrete Applied Mathematics **155** (2007), 1254–1266. https://doi.org/10.1016/j.dam.2006.10.014.

[5] Fangxu Xi and Ligong Wang, *Integral trees with diameter 6*, Discrete Applied Mathematics **358** (2024), 76–90. https://doi.org/10.1016/j.dam.2024.06.039.

[6] Robert A. Beezer, *Trees with very few eigenvalues*, Journal of Graph Theory **14** (1990), 509–517. https://doi.org/10.1002/jgt.3190140415.